

Ayush Raj

Third Year Undergraduate
Department of Aerospace Engineering

✉ ayushraj23@iitk.ac.in | 📞 +91-6901768815
🌐 Ayush Raj | 📧 Ayushraj6942

Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2023-Present	B.Tech	Indian Institute of Technology Kanpur	6.6/10
2023	CBSE(XII)	Indian Convent School,Kota	94.2%
2021	CBSE(X)	Army Public School,Sri Ganganagar	96.4%

Scholastic Achievements

- Secured All India Rank **3946** in **JEE Advanced 2023** among 2.5 lakh shortlisted candidates.
- Secured All India Rank **4175** in **JEE Mains 2023** among 11 lakh candidates.

Research Experience

SURGE’25 IIT KANPUR | B2B Brand Equity | Prof.RN Sengupta (May’25-July’25)

Objective	• Identify determinants of B2B brand equity using literature, company data, and test models via ML .
Approach	• Reviewed literature on B2B Brand Equity to map constructs like awareness, loyalty, trust, and reputation. • Applied SEM and multiple logistic regression on 5 firms; analyzed impact of digital marketing using MROI . •Analysed Semantic Brand Score to measure brand importance and strength using textual data and Monte Carlo Simulation for Brand Equity Risk Optimization
Impact	• Proposed methods to quantify brand equity using ML models and highlighted digital marketing’s role via MROI .

Key Projects

Financial Markets & Investment Banking | Finlatics 📊 (Dec’24-Feb’25)

- Managed equity portfolio on **BSE 500 Index**; applied portfolio construction diversification, scored **Finscore: 81.16**.
- Conducted sectoral analysis using **Porter’s 5 Forces**; portfolio restructuring and strategic fund positioning; rated **82.12**.
- Completed **Private Equity** and **Investment Banking** project covering investments, fund setup case studies; rated **91.21**.

Hydro Mechanics Of Finvision | Google Developer Group Project| 📊 (Dec’24-Mar’25)

- Built real-time financial data pipelines via **Selenium, Requests, BeautifulSoup** from **Yahoo Finance, NSE**, etc.
- Applied **Black-Scholes, Binomial, Vasicek, CIR, Merton Models**; simulated **GBM, ADF, Weiner** processes for forecasting & risk.
- Used **ML (LSTM, RNN)** for **ETH** prediction; explored quantum models using **TensorFlow Quantum, PennyLane**.

Course Project-DMS624: Computer-Aided Decision Systems | Prof. Deepu Philip 📊 (Aug’25-Nov’25)

- OrbitWear: End-to-End Merchandise Decision System**
 - Designed **end-to-end merchandise solution** (design, pricing, delivery) for IITK clubs, resolving vendor coordination issues.
 - Managed **pricing, advance payments, vendor payouts, and profit tracking** under a low-capital model to minimize financial risk.
- Solving Farmers’ Loan Access: Digital Kisan ID–Based Credit Framework**
 - Analyzed credit access barriers using **NABARD, RBI, & PM-KISAN reports** to identify systemic bottlenecks.
 - Proposed **Digital Kisan ID** integrating Aadhaar & land records to **auto-fill 80% of loan forms** for instant application.

Technical Skills

- Programming Languages:** C, C++, Python • **Software:** Canva, MS Excel, Jupyter
- Libraries:** Numpy, Pandas, Tensorflow, Seaborn, Matplotlib, Keras, Scikit-Learn

Relevant Courses

Introduction To Management	Fundamentals of Computing	Application of Single Variable Calculus
Complex Variable	Linear Algebra	Ordinary Differential Equations

Positions of Responsibility

Business Head | IITK Rocketry & Space Exploration Team Mar’24-Present

- Contributed to **avionics, structures, propulsion, recovery**; launched rocket to **600 ft** apogee in **July 2024**.
- Managed finances; secured **2 sponsorships**, handled **5 lakh** budget, and oversaw bill clearance.
- Led **30+ juniors**, handled outreach, sponsorship, and participating as Recovery member in **INSPACE Competition**.

Marketing Secretary | Udghosh’24 Aug’24-Oct’24

- Led lead generation and **database management** for **100+ companies** to support sponsorship and marketing outreach.
- Developed strong **communication skills** including cold calling, email writing, and persuasive brand negotiation.

Extra-Curricular Activities

- Received **2 LORs and certificates** for outstanding performance in **finlatics financial markets & investment banking** project.
- Received **certificate** for participating in **IIT Kanpur certification program on Python For AI,ML and DL**.
- Completed **NCC ‘C’ Certificate** program, demonstrating leadership, discipline, and endurance.
- Cleared **PDR and CDR** rounds of **INSPACE Model Rocketry Competition** with IITK RaSET team.